DATA MANAGEMENT FOR MACHINE LEARNING

Assignment 2: End-to-End Data Management Pipeline for a Recommendation System

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Project	E-commerce Recommendation System
Environment	Google Colab
Programming Language	Python

1. Project Objective

The objective of this project is to design and implement an end-to-end data management pipeline for an e-commerce recommendation system. The pipeline covers data ingestion, validation, cleaning, preparation, feature engineering, recommendation model training, evaluation and workflow orchestration.

2. Dataset Description

Dataset	Records	Columns
Users	500	4
Items	200	4
Interactions	12,000	6

Synthetic data was generated for 500 users, 200 products and 12,000 user-item interactions. The interaction records contain ratings, interaction types and timestamps.

3. Data Validation and Preparation

The datasets were checked for missing values, duplicate records, invalid ratings, invalid interaction types, invalid user and item references, invalid timestamps and invalid item prices. All 12 validation checks passed with zero invalid records. The cleaned datasets preserved all 500 users, 200 items and 12,000 interactions. An 80:20 chronological split produced 9,600 training records and 2,400 testing records.

4. Feature Engineering

User features include interaction count, average rating and preferred category. Item features include interaction count, average rating and popularity score. Interaction features include days since interaction and product category. The feature catalog was saved as a CSV file.

5. Recommendation Models

Model 1 – Popularity-Based Recommendation: Ranks products according to their interaction counts in the training dataset and recommends globally popular products.

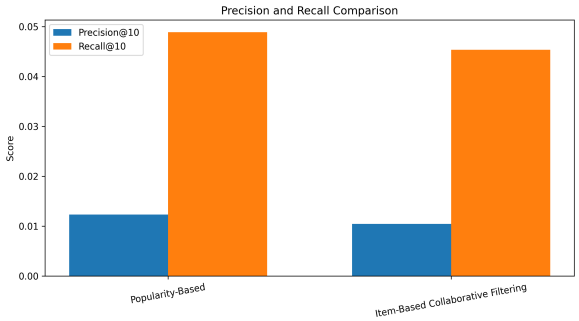
Model 2 – Item-Based Collaborative Filtering: Uses a user-item rating matrix and cosine similarity between items to generate personalized recommendations.

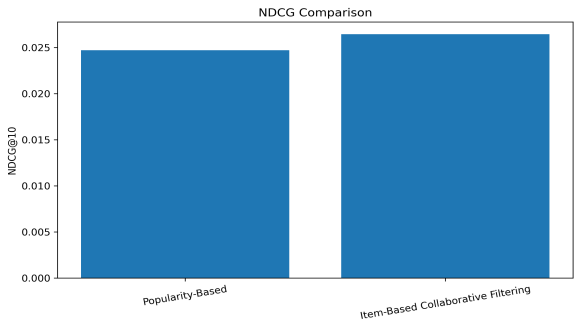
6. Model Evaluation

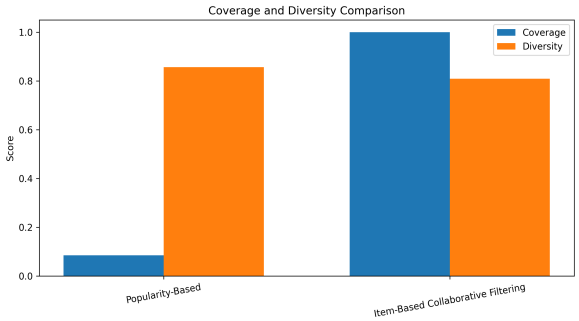
Metric	Popularity	Collaborative Filtering
Precision@10	0.012297	0.010441
Recall@10	0.048884	0.045343
NDCG@10	0.024728	0.026468
Coverage	0.085	1.000
Diversity	0.856458	0.809126

Evaluation used test interactions with ratings of 4 or 5 as relevant items. Both models were evaluated using the same 431 eligible users and top-10 recommendations.

7. Evaluation Visualizations







8. Workflow Orchestration

A dependency-ordered Python workflow was developed to verify the output files of each pipeline stage. The workflow checked data ingestion, validation, preparation, feature engineering, model training and model evaluation. All six stages reported SUCCESS, with no missing files.

9. System Architecture

Raw Data Sources → Data Ingestion → Data Validation → Data Cleaning and Preparation → Feature Engineering → Chronological Train/Test Split → Popularity-Based Model and Collaborative Filtering → Model Evaluation → Workflow Orchestration → Final Reports

10. Results and Discussion

The popularity-based model achieved higher Precision@10, Recall@10 and Diversity. The collaborative filtering model achieved higher NDCG@10 and full catalog coverage. Therefore, neither model was best across every metric. The experiment uses synthetic data, so the measured scores should be interpreted as results of this demonstration rather than evidence of real-world recommendation quality.

11. Conclusion

An end-to-end data management pipeline for an e-commerce recommendation system was implemented successfully. The project demonstrated data ingestion, validation, preparation, feature engineering, chronological data splitting, two recommendation approaches, model evaluation and workflow execution. All six workflow verification stages completed successfully.

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